

Smart Home Appliance Energy Consumption Prediction

IoT & Data Analytics Course Project

Team Name
MECHMASTERS

Team Members

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Problem Statement

Why this project?



Smart homes generate large amounts of sensor data



Energy consumption is costly and inefficient when unmanaged



Predicting usage helps optimize energy behavior

Problem Statement

Predict appliance energy consumption using environmental and weather sensor data.



Dataset Overview

Dataset: Appliances Energy Prediction (UCI / Kaggle)

Contains

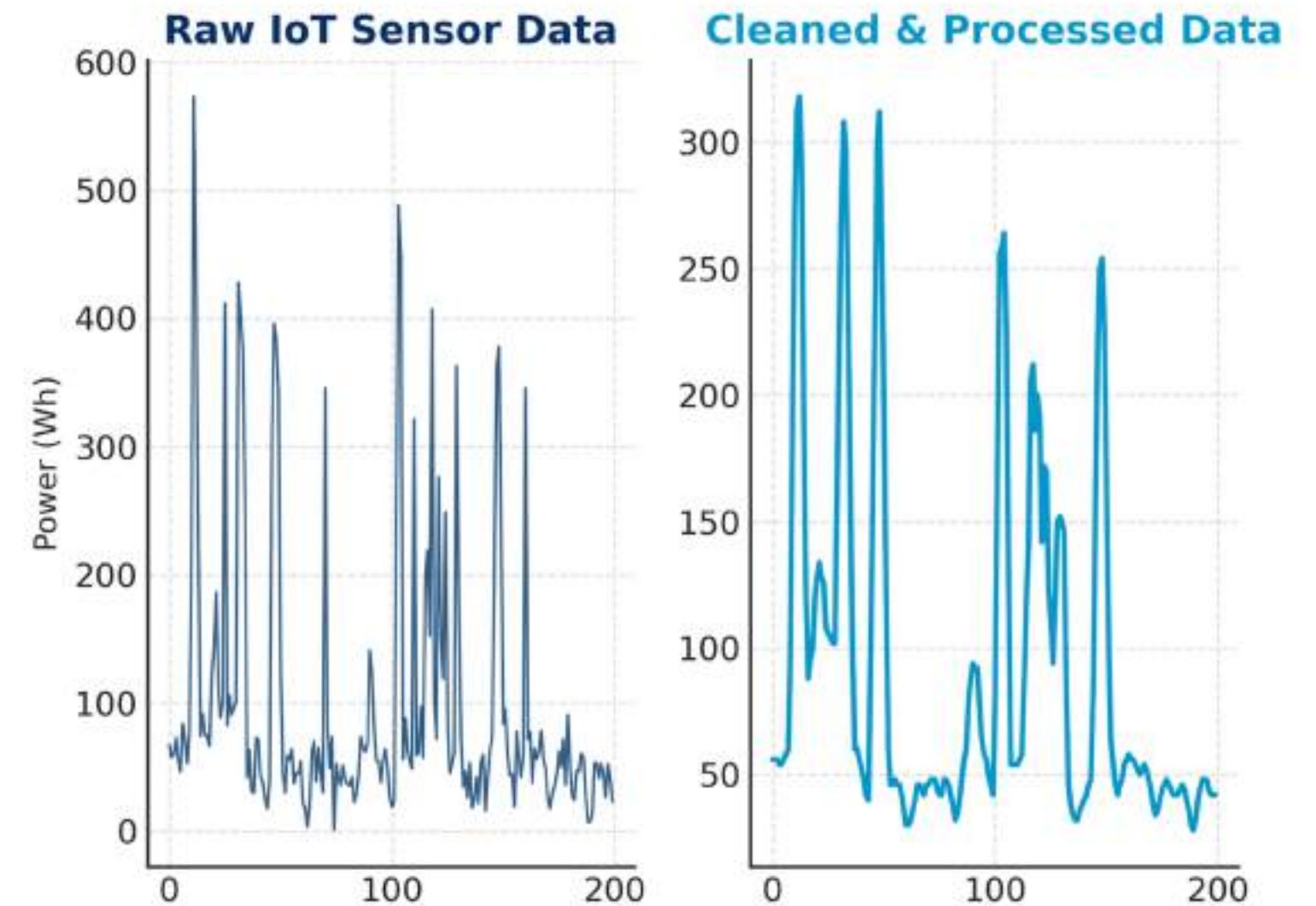
- Indoor temperature & humidity (multiple rooms)
- Outdoor weather data
- Time-based information
- Energy usage (target variable)

Frequency

Every 10 minutes

Duration

~4.5 months



Data Preparation (Google Colab)

Tools Used

- Google Colab
- Python (Pandas, NumPy, Scikit-learn)

Steps

- Loaded CSV data
- Converted date to datetime
- Extracted time features (hour, day, month, week)
- Removed irrelevant variables (rv1, rv2)



What we analyzed

Outcome

A stylized illustration of a person with dark hair, wearing a white long-sleeved shirt and white pants, sitting at a black desk and working on a computer. The person is smiling and looking at the monitor. On the desk, there is a large white monitor and a keyboard. To the left of the person, there is a white server rack with four drawers. Above the person, there are several floating icons: a large white gear, a magnifying glass over a bar chart, a line graph with an upward arrow, a pie chart, and a megaphone. The background is a solid blue color.

Feature Scaling & Data Split

Preprocessing



**Train-test split
(80% / 20%)**



**Standardization
using
StandardScaler**



Why scaling matters



**Ensures all features
contribute fairly**



**Improves model
performance**

Model Training

Models Tested:



• **Linear Regression**



• **Decision Tree**



• **Random Forest**



• **Gradient Boosting**



Evaluation Metrics:



• **MAE (Mean Absolute Error)**



• **RMSE (Root Mean Squared Error)**



Model Selection

Best Model:

- **Random Forest Regressor**

Why?

- **Lowest prediction error**
- **Handles non-linear relationships**

- **Robust to noise**

Saved artifacts:

- **Trained model (.pkl)**
- **Scaler (scaler.pkl)**

Dashboard (Local Deployment)

Technology:

- 🔍 Streamlit (Python-based dashboard)

Features:

- 🔍 User inputs sensor values
- 🔍 Data is scaled using saved scaler
- 🔍 Model predicts energy consumption instantly



Smart Home Energy Prediction Dashboard

Enter the environmental sensor values below to predict appliance energy consumption.

Enter lights: 30.00 - +

Enter T1: 9.89 - +

Enter RH_1: 6.77 - +

Enter T2: 19.20 - +



What is AWS

Amazon Web Services (AWS) is a cloud computing platform by Amazon. It provides virtual servers and services without physical hardware.





Why Use Cloud Computing

- No physical hardware
Everything runs on virtual servers.
- Access from anywhere
The system can be used from any location.
- Scalable and reliable
Resources can grow based on needs

AWS in Our Project

- **Cloud-hosted dashboard**

The dashboard was deployed on the cloud to allow remote access through the internet.

- **AWS EC2 service**

An EC2 instance was used as a virtual server to host and run the application.

- **Ubuntu Linux operating system**

The EC2 server runs on Ubuntu, which is a Linux-based operating system used to execute all commands and applications.



What is EC2?

- Virtual machine
Acts like a real computer in the cloud.
- Always running
Available 24/7.
- Customizable
OS, security, and performance can be selected.



Amazon
EC2





Creating the EC2 Instance

An EC2 instance was created using the AWS Management Console with the following configuration:

- AMI: Ubuntu (64-bit)
- Instance Type: t2.micro
- Key Pair: A private key file (key.pem) for SSH authentication
- Security Group Rules:
 - Port 22 (SSH) – for remote access
 - Port 8501 – for accessing the Streamlit dashboard

After launching the instance, AWS assigned a public IPv4 address, which was later used to access the dashboard from a web browser.

The screenshot shows the AWS Management Console interface for the EC2 Instances page. The top navigation bar includes the AWS logo, a search bar, and the region 'Europe (Frankfurt)'. The main content area displays a table of instances with columns for Name, Instance ID, Instance state, Instance type, Status check, Alarm status, Availability Zone, and Public IPv4 DNS. A single instance named 'smart house' with ID 'i-0daeca447517ab122' is listed in a 'Running' state. Below the table, the 'Details' tab is selected, showing the 'Instance summary' for the selected instance. The summary includes the Instance ID, Public IPv4 address (3.123.230.138), Private IPv4 address (172.31.8.222), Instance state (Running), Instance type (t3.micro), and other configuration details.



Dashboard Deployment



- Built with Streamlit
Used to create the web dashboard.



- Runs on AWS server
Executed remotely on EC2.



- Python-based application
Handles data and predictions.



Windows PowerShell

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Install the latest PowerShell for new features and improvements! <https://aka.ms/PSWindows>

PS C:\Users\LENOVO> cd Desktop

PS C:\Users\LENOVO\Desktop> ssh -i key.pem ubuntu@3.123.230.138

Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1015-aws x86_64)

- * Documentation: <https://help.ubuntu.com>
- * Management: <https://landscape.canonical.com>
- * Support: <https://ubuntu.com/pro>

System information as of Thu Dec 25 01:17:21 UTC 2025

System load:	0.04	Temperature:	-273.1 C
Usage of /:	26.9% of 14.46GB	Processes:	115
Memory usage:	27%	Users logged in:	0
Swap usage:	0%	IPv4 address for ens5:	172.31.8.222

- * Ubuntu Pro delivers the most comprehensive open source security and compliance features.

<https://ubuntu.com/aws/pro>

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See <https://ubuntu.com/esm> or run: `sudo pro status`

*** System restart required ***

Last login: Thu Dec 25 01:14:56 2025 from 85.100.64.96

```
*** System restart required ***
Last login: Thu Dec 25 01:14:56 2025 from 85.100.64.96
ubuntu@ip-172-31-8-222:~$ sudo apt update && sudo apt upgrade -y
Hit:1 http://eu-central-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://eu-central-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://eu-central-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
ubuntu@ip-172-31-8-222:~$ python3 -m venv venv
source venv/bin/activate
(venv) ubuntu@ip-172-31-8-222:~$ pip install streamlit pandas numpy scikit-learn
Requirement already satisfied: streamlit in ./venv/lib/python3.12/site-packages (1.52.1)
Requirement already satisfied: pandas in ./venv/lib/python3.12/site-packages (2.3.3)
Requirement already satisfied: numpy in ./venv/lib/python3.12/site-packages (2.3.5)
Requirement already satisfied: scikit-learn in ./venv/lib/python3.12/site-packages (1.6.1)
Requirement already satisfied: altair!=5.4.0,!=5.4.1,<7,>=4.0 in ./venv/lib/python3.12/site-packages (from streamlit) (6.0.0)
Requirement already satisfied: blinker<2,>=1.5.0 in ./venv/lib/python3.12/site-packages (from streamlit) (1.9.0)
Requirement already satisfied: cachetools<7,>=4.0 in ./venv/lib/python3.12/site-packages (from streamlit) (6.2.4)
Requirement already satisfied: click<9,>=7.0 in ./venv/lib/python3.12/site-packages (from streamlit) (8.3.1)
Requirement already satisfied: packaging>=20 in ./venv/lib/python3.12/site-packages (from streamlit) (25.0)
Requirement already satisfied: pillow<13,>=7.1.0 in ./venv/lib/python3.12/site-packages (from streamlit) (12.0.0)
Requirement already satisfied: protobuf<7,>=3.20 in ./venv/lib/python3.12/site-packages (from streamlit) (6.33.2)
Requirement already satisfied: pyarrow>=7.0 in ./venv/lib/python3.12/site-packages (from streamlit) (22.0.0)
Requirement already satisfied: requests<3,>=2.27 in ./venv/lib/python3.12/site-packages (from streamlit) (2.32.5)
```

```
Requirement already satisfied: MarkupSafe>=2.0 in ./venv/lib/python3.12/site-packages (from jinja2->alt
3)
Requirement already satisfied: attrs>=22.2.0 in ./venv/lib/python3.12/site-packages (from jsonschema>=3
) (25.4.0)
Requirement already satisfied: jsonschema-specifications>=2023.03.6 in ./venv/lib/python3.12/site-packa
4.1,<7,>=4.0->streamlit) (2025.9.1)
Requirement already satisfied: referencing>=0.28.4 in ./venv/lib/python3.12/site-packages (from jsonsch
eamlit) (0.37.0)
Requirement already satisfied: rpds-py>=0.7.1 in ./venv/lib/python3.12/site-packages (from jsonschema>=
t) (0.30.0)
(venv) ubuntu@ip-172-31-8-222:~$ streamlit run app.py --server.port 8501 --server.address 0.0.0.0

Collecting usage statistics. To deactivate, set browser.gatherUsageStats to false.

You can now view your Streamlit app in your browser.

URL: http://0.0.0.0:8501
```

- After running the command, the dashboard service started successfully on the server. The application was then accessed from a local machine using a web browser by entering the public IP address of the EC2 instance followed by the port number:

<http://3.123.230.138:8501>

Accessing the Dashboard

- **Public IP address**

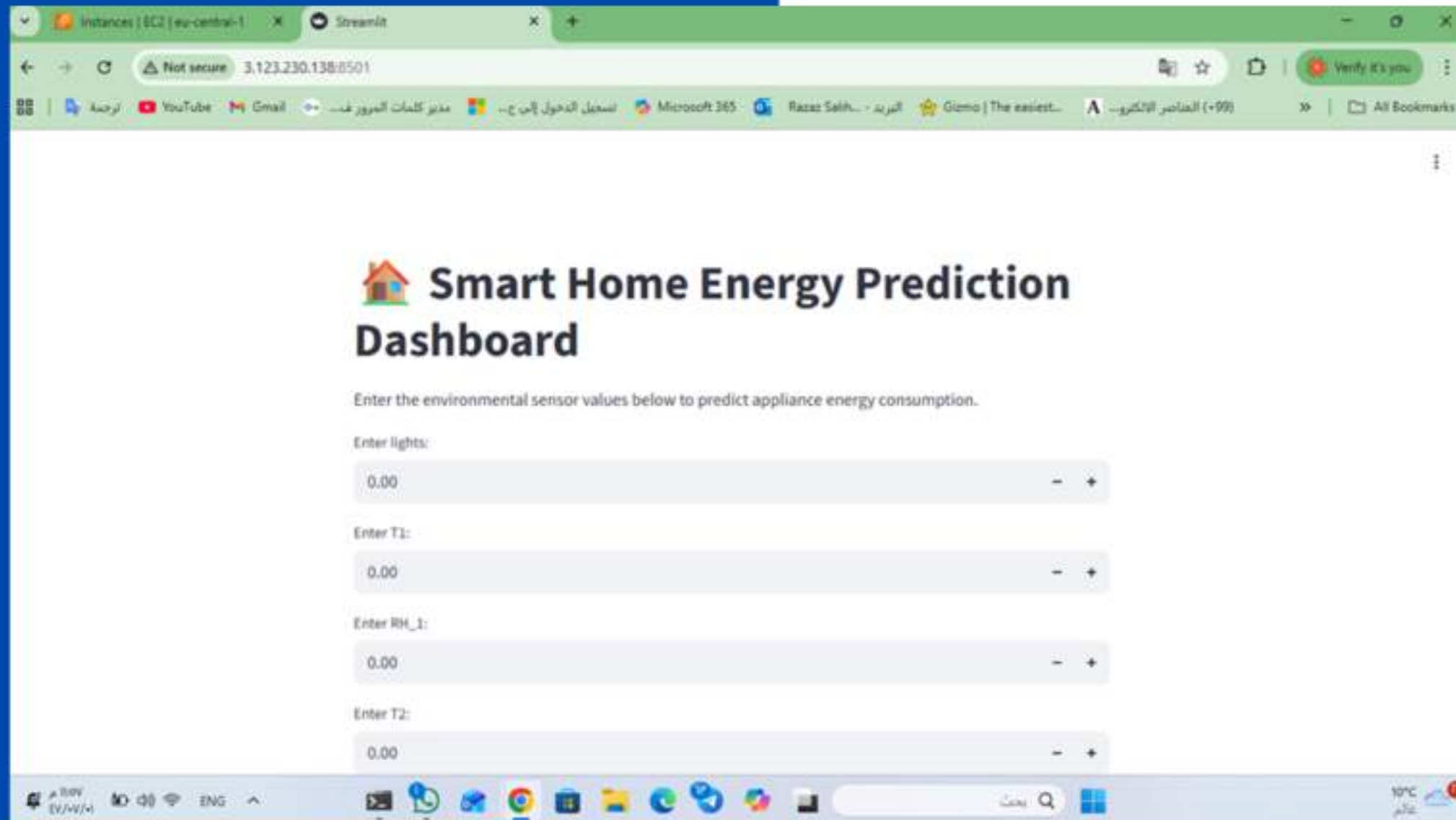
Used to reach the application.

- **Web browser access**

No installation required for users.

- **Real-time results**

Outputs are displayed instantly.



Enter hour:

0.01

- +

Enter day_of_week:

0.01

- +

Enter month:

0.01

- +

Enter week_of_year:

0.01

- +

Predict Energy Consumption

Predicted Appliance Energy Use: 40.85 Wh



**Thank You
Very Much!**

